



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (2U)

2017 HSC Assessment Task 1

Wednesday 7th December

General Instructions

- Working time – 55 minutes (plus 5 minutes reading time).
- Use a blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.
- Commence each question on a new page.
- Write on both sides of the paper.
- Show all necessary working for every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: ...4.....

BOOKLETS USED:

Class (please ✓)

☐

Mr Berry

☐

Ms Lee

☐

Mr Hwang

☐

Mr Lin

☐

Mr Ireland

☐

Ms Ziazaris

☐

Dr Jomaa

☐

Mr Zuber

Question	1 Series & Sequences	2 Locus and the Parabola	3 Quadratic Theory	4 Challenge	Total	%
Marks	<u>14</u>	<u>9</u>	<u>13</u>	<u>3</u>	<u>39</u>	<u>100</u>

Question 1 (14 marks)**Marks**

- (a) Find the 47th term in the arithmetic sequence $-4, -1, 2, 5, \dots$ **2**

- (b) Find the sum of the first 10 terms of the following geometric series: **2**

$$16 - 48 + 144 - 432 + \dots$$

- (c) The 8th term of an arithmetic series is 16 and the sum of the first 10 terms is 210. **4**

Find the sum of the first 15 terms.

- (d) Determine the value of n if **3**

$$\sum_{p=1}^n 2p - 3 = 80$$

- (e) The following is a geometric series:

$$1 + \frac{1}{1 + \sqrt{2}} + (3 - 2\sqrt{2}) + \dots$$

- i) Explain why this geometric series has a limiting sum. **1**

- ii) Find the limiting sum in the form $a + b\sqrt{c}$. **2**

Examination continues on the next page.

Question 2 (9 marks)

- (a) For the equation $12x + y^2 + 4y = 44$ find:
- i) The vertex. 2
 - ii) The focus. 1
 - iii) The equation of the directrix. 1
- (b) Given the points $A(4, 1)$ and $B(2, -3)$ and the point $P(x, y)$
- i) Write an expression for the gradient PA . 1
 - ii) Hence or otherwise find the equation of the locus such that $\angle APB$ is a right angle. 2
 - iii) Describe the locus geometrically. 2

Examination continues on the next page 

Question 3 (13 marks)

- (a) Find the values of k for which equation $2x^2 - 5x - k = 0$ has real roots. 3
- (b) If α and β are the roots of the quadratic equation $x^2 - 3x - 5 = 0$, without solving the equation, find the values of :
- (i) $\alpha + \beta$ 1
 - (ii) $\alpha\beta$ 1
 - (iii) $(\alpha - 2)(\beta - 2)$ 1
 - (iv) $\alpha^2 + \beta^2$ 1
 - (v) $|\alpha - \beta|$ 1
- (c) Given the quadratic equation $x^2 - (2 + m)x + 3m = 0$, find the value of m if the product of the roots is four times the sum of the roots. 2
- (d) Solve for x : 3
- $$(x + 8)^4 = (2x + 16)^2$$

Question 4 (3 marks)

- (a) Find the sum of the roots of the equation 3

$$(x + 1)(x + 2) + (x + 3)(x + 4) + \cdots + (x + 99)(x + 100) = 0$$

End of examination.

Solutions NSBHS Mathematics (24) 2017 Task 1

Q1 a) $T_1 = -4, T_2 = -1, T_3 = 2$
AP with $a = -4, d = 3$
 $T_{47} = -4 + (47-1) \times 3$
 $= 134.$

b) $S_n = 16 - 48 + 144 - 432 + \dots$
is a GP with $a = 16, r = -3$

$$\begin{aligned}\therefore S_{10} &= \frac{a(r^{10}-1)}{r-1} \\ &= \frac{16((-3)^{10}-1)}{(-3)-1} \\ &= -236,192\end{aligned}$$

c) AP $T_8 = 16 = a + 7d$ (1)
 $S_{10} = 210 = \frac{10}{2}(2a + 9d)$ (2)

$$\begin{cases} a + 7d = 16 & (1) \\ 2a + 14d = 42 & (2) \end{cases}$$

$$2a + 7d = 32 \quad (1) \times 2$$

$$2a + 14d = 42$$

$$5d = -10$$

$$d = -2$$

$$\begin{aligned}\therefore a &= 16 - 7(-2) \\ &= 30\end{aligned}$$

$$\therefore S_{15} = \frac{15}{2}(2(30) + 14(-2)) = 240$$

$$\text{Q/d. } \sum_{p=1}^n 2p-3=80$$

This is an AP, $T_1=2(1)-3=-1$
 $T_2=2(2)-3=-3$
 $T_3=2(3)-3=-3$

$$\therefore S_n = \frac{n}{2}(a+l)$$

$$= \frac{n}{2}((-1) + (2n-3))$$

$$\therefore \frac{n}{2}(2n-4)=80$$

$$n^2 - 2n - 80 = 0$$

$$(n+8)(n-10) = 0$$

$$n=10 \quad (n>10)$$

e) GP: $1 + \frac{1}{1+\sqrt{2}} + (3-2\sqrt{2}) + \dots$

(i) We are told this is a GP $\therefore r = \frac{1}{1+\sqrt{2}} = \frac{1}{1+\sqrt{2}}$

$$1+\sqrt{2} > 1$$

$$\therefore \frac{1}{1+\sqrt{2}} < 1 \quad \text{and} \quad \frac{1}{1+\sqrt{2}} > 0$$

$$\therefore |r| < 1 \quad \text{so we have a limiting sum.}$$

OR $\frac{1}{1+\sqrt{2}} \approx 0.4142 \Rightarrow |r| < 1$

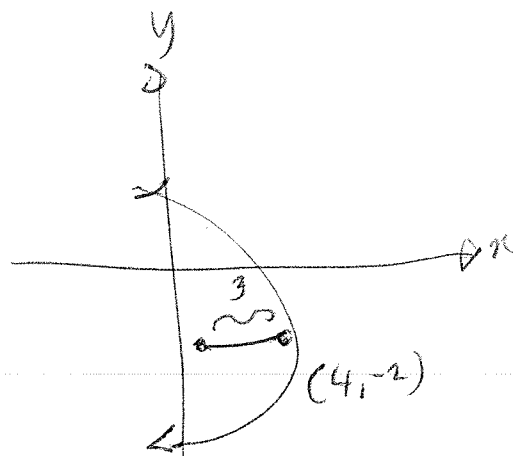
$$(ii) S_{\infty} = \frac{a}{1-r} = \frac{1}{1 - \frac{1}{1+\sqrt{2}}} = \frac{1}{\frac{1+\sqrt{2}-1}{1+\sqrt{2}}} = \frac{1+\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}+2}{2} = 1 + \frac{1}{2}\sqrt{2}$$

$$\begin{aligned}
 \text{Q2 (a)} \quad 12x + y^2 + 4y &= 44 \\
 y^2 + 4y &= -12x + 44 \\
 (y+2)^2 - 4 &= -12x + 44 \\
 (y+2)^2 &= -12x + 48 \\
 (y+2)^2 &= -12(x-4)
 \end{aligned}$$

$\therefore$ Vertex is $(4, -2)$
 $a = 3$

Focus is $(1, -2)$

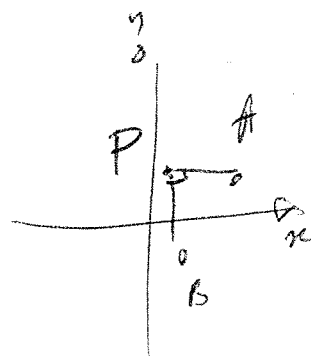
Directrix $x = 7$



(b) Let $P(x, y)$ be on the locus.

$$(i) \quad m_{PA} = \frac{y-1}{x-4}$$

$$\begin{aligned}
 A(4, 1) \\
 B(2, -3)
 \end{aligned}$$



$$(ii) \quad m_{PB} = \frac{y+3}{x-2}$$

$$\text{Require } m_{PA} = -\frac{1}{m_{PB}}$$

$$i. \quad \frac{y-1}{x-4} = -\left(\frac{x-2}{y+3}\right)$$

$$(y-1)(y+3) = -(x-2)(x-4)$$

$$y^2 + 2y - 3 = -(x^2 - 6x + 8)$$

$$x^2 - 6x + 8 + y^2 + 2y - 3 = 0$$

$$\begin{aligned}
 (x-3)^2 - 9 + 8 + (y+1)^2 - 1 - 3 &= 0 \\
 (x-3)^2 + (y+1)^2 &= 5
 \end{aligned}$$

Locus is a
 Circle
 Centre $(3, -1)$
 radius $\sqrt{5}$

$$Q3a) \quad 2x^2 - 5x - k = 0$$

has real roots when $\Delta \geq 0$

$$(-5)^2 - 4(2)(-k) \geq 0$$

$$25 + 8k \geq 0$$

$$k \geq -\frac{25}{8}$$

$$b) \quad x^2 - 3x - 5 = 0$$

$$i) \alpha + \beta = 3$$

$$ii) \alpha\beta = -5$$

$$\begin{aligned} iii) (\alpha - 2)(\beta - 2) &= \alpha\beta - 2(\alpha + \beta) + 4 \\ &= -5 - 2(3) + 4 \\ &= -7 \end{aligned}$$

$$\begin{aligned} iv) \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= (3)^2 - 2(-5) \\ &= 9 + 10 \\ &= 19 \end{aligned}$$

$$\begin{aligned} v) |\alpha - \beta| &= \sqrt{(\alpha - \beta)^2} \\ &= \sqrt{\alpha^2 + \beta^2 - 2\alpha\beta} \\ &= \sqrt{19 - 2(-5)} \quad \text{from (iv)} \\ &= \sqrt{29} \end{aligned}$$

OR use $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

OR $\alpha = \frac{-b + \sqrt{\Delta}}{2a}$

$$\beta = \frac{-b - \sqrt{\Delta}}{2a}$$

$$\therefore |\alpha - \beta| = \frac{2\sqrt{\Delta}}{2a} = \frac{\sqrt{\Delta}}{a}$$

$$Q3 (a) \quad x^2 - (24m)x + 3m = 0$$

$$\alpha\beta = 4(\alpha + \beta)$$

$$3m = 4(24m)$$

$$3m = 8 + 4m$$

$$m = -8$$

$$(d) \quad (x+8)^4 = (2x+16)^2$$

$$(x+8)^4 = (2(x+8))^2$$

$$(x+8)^4 = 4(x+8)^2$$

$$\text{Let } u = (x+8)^2$$

$$u^2 = 4u$$

$$u^2 - 4u = 0$$

$$u(u-4) = 0$$

$$\therefore x+8=0 \text{ or } (x+8)^2=4$$

$$x+8 = \pm 2$$

$$x = -8$$

$$\therefore x = -6 \text{ or } x = -10$$

$$\therefore x = -10, -8, -6$$

Q4.

$$(x+1)(x+2) + (x+3)(x+4) + \dots + (x+99)(x+100) = 0$$

$$x^2 + 3x + 2 + x^2 + 7x + 12 + \dots + x^2 + 199x + 9900 = 0$$

$$50x^2 + (3+7+\dots+199)x + K = 0$$

AP: 50 terms

$$\text{b coefficient} = \frac{50}{2}(3+199) = 5050$$

$$\therefore \alpha + \beta = -\frac{b}{a} = \frac{-5050}{50} = -101$$